

Fundamentals of ICT Part A (continued): Network Fundamentals and Internet Evolution

I. Key Terms Associated with Networks A computer network is not just about connecting wires; it is about establishing a shared language between machines so they can exchange resources (like printers or files) and communicate. To understand how the internet works, you must understand the vocabulary of networking.

- **Node:** In network terminology, any device connected to the network is called a "node." This isn't limited to computers; your smartphone, a smart lightbulb, a network printer, or a server are all nodes. If it has an address and can send/receive data, it is a node.
- **Packet:** Imagine sending a 500-page book to a friend by mailing one page at a time in separate envelopes. That is how the internet works. A large file (like a video) is broken down into tiny units called **Packets**. These packets travel independently across the web—taking different routes to avoid traffic—and are reassembled into the correct order once they reach the destination.
- **Protocol:** Computers need strict rules to understand each other. A **Protocol** is a set of rules or standards that govern how data is transferred. It is the "etiquette" of the network. For example, if one computer speaks "English" and the other speaks "French," they cannot communicate. Protocols ensure they both speak the same language (e.g., HTTP for websites, SMTP for email).
- **Bandwidth vs. Latency:** These are the two measures of network speed, but they mean different things.

Bandwidth is the *width* of the pipe. It measures how *much* data can be sent at once (e.g., 100 Mbps). High bandwidth is good for downloading large movies.

Latency (Ping) is the *length* of the pipe. It measures how *fast* a single signal can travel from you to the server and back. Low latency is crucial for online gaming or video calls, where a delay of even 100 milliseconds causes "lag."

- **IP Address vs. MAC Address:** Every device needs an ID card.

The **IP Address (Internet Protocol)** is your "virtual address." It tells the network *where* you are located effectively (like a street address). It can change depending on which Wi-Fi you connect to.

The **MAC Address (Media Access Control)** is your "physical address." It is burnt into the network card at the factory and never changes. It identifies *who* the device is (like a Social Security Number).

- **Network Architectures (Client-Server vs. P2P):**

Client-Server: This is the standard model of the web. A powerful central computer (the **Server**) holds the files, and many smaller computers (**Clients**) request to see them. When you watch Netflix, you are the client requesting a movie from Netflix's server.

Peer-to-Peer (P2P): In this model, there is no boss. All computers are equal and share files directly with each other. This is how BitTorrent works—you download parts of a file from 50 different people's computers simultaneously, not from one central server.

II. Milestones in the Development of the Internet The Internet did not appear overnight. It evolved from a military experiment into the global utility we use today.

- **The 1960s: The Cold War Roots (ARPANET)** The internet was born out of fear. During the Cold War, the US government worried that a nuclear attack could destroy their centralized communication lines. In **1969**, the Department of Defense created **ARPANET**. It was the first "decentralized" network, designed so that if one node was destroyed, the data could "reroute" around the damage. The first message sent was intended to be "LOGIN," but the system crashed after the first two letters, sending only "LO."
- **The 1970s: Teaching Computers to Talk (TCP/IP)** In the early 70s, different networks existed, but they couldn't talk to each other (like having an AT&T phone that couldn't call a Verizon phone). In **1974**, Vint Cerf and Bob Kahn invented **TCP/IP** (Transmission Control Protocol/Internet Protocol). This technology acted as a universal translator, allowing different types of networks to connect. This "inter-networking" is where the word "Internet" comes from.
- **The 1980s: Organizing the Chaos (DNS)** Originally, to visit a website, you had to type its numerical IP address (e.g., 192.168.0.1). This was impossible to memorize. In **1984**, the **Domain Name System (DNS)** was introduced. It acted as the "phonebook" of the internet, matching human-readable names (like "https://www.google.com/search?q=google.com") to the machine-readable IP addresses.
- **The 1990s: The Web for Everyone (WWW)** Before 1990, the internet was just black screens with green text, used only by scientists. In **1991**, Tim Berners-Lee, a scientist at CERN, invented the **World Wide Web (WWW)**. He created **HTML** (to format pages) and **HTTP** (to send them).

Critical Distinction: The **Internet** is the physical cables and computers. The **World Wide Web** is the pages and links we see on top of it.

In **1993**, **Mosaic** was released. It was the first web browser that could display images inline with text, making the web visually appealing to regular people for the first time.

- **The 2000s: The Social and Mobile Web (Web 2.0)** Around 2004, the internet shifted from "Read-Only" (Web 1.0) to "Read-Write" (Web 2.0). Users stopped just consuming content and started creating it via blogs, YouTube, and Facebook.

In **2007**, the launch of the **iPhone** changed the internet again. It moved connectivity from the desk to the pocket, creating an "always-on" culture where users are connected 24/7.

- **The 2010s–Present: The Internet of Things (IoT)** Today, the internet has expanded beyond computers. The **Internet of Things (IoT)** refers to everyday objects—refrigerators, cars, doorbells—connecting to the network. With the rollout of **5G**, these devices can communicate instantly, paving the way for self-driving cars and smart cities.